MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 05 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 12: Surface Areas and Volumes** | **MM:** 24

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.
3. Total marks of printed questions Q1–Q11 $= 5 + 4 + 6 + 5 + 4 = 24$ Marks (matching section totals; source paper header prints MM: 30).

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | $(5 \times 1 = 5$ Marks$)$

Q1. The edge of a cube whose volume is equal to that of a cuboid of dimensions 8 cm × 4 cm × 2 cm is: [1]

(A) 6 cm (B) 4 cm (C) 2 cm (D) 8 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 4 cm** [0.5 Mark]

Formula: Volume of Cube $a^3 = lbh = 8 \times 4 \times 2 = 64 = 4^3 \implies$ **a = 4 cm**.

Q2. Three solid metal cubes of edges 6 cm, 8 cm, and 10 cm are melted to form a single cube. The length of the diagonal of the larger cube formed is: [1]

(A) $4\sqrt{6}$ cm (B) $15\sqrt{2}$ cm (C) $12\sqrt{3}$ cm (D) $18\sqrt{3}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) $12\sqrt{3}$ cm** [0.5 Mark]

Given: $A^3 = 6^3 + 8^3 + 10^3 = 216 + 512 + 1000 = 1728 = 12^3 \implies A = 12$ cm.

Formula / Theorem Used: Diagonal of Cube $= A\sqrt{3} =$ **$12\sqrt{3}$ cm**

Q3. Three solid cubes have a face diagonal of $4\sqrt{2}$ cm each. Three other solid cubes have a face diagonal of $8\sqrt{2}$ cm each. All six are melted to form a single cube. The edge length is: [1]

(A) 12 cm (B) 24 cm (C) 18 cm (D) 16 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) 12 cm** [0.5 Mark]

Given: Face diagonal $= a\sqrt{2} \implies a_1\sqrt{2} = 4\sqrt{2} \implies a_1 = 4$ cm and $a_2 = 8$ cm.

Formula / Theorem Used: $A^3 = 3a_1^3 + 3a_2^3 = 3(4^3 + 8^3) = 3(64 + 512) = 3 \times 576 = 1728 \implies$ **A = 12 cm**

Q4. The sum of length, breadth, and height of a cuboid is $6\sqrt{3}$ cm and diagonal is $2\sqrt{6}$ cm. The total surface area of the cuboid is: [1]

(A) 48 cm^2 (B) 72 cm^2 (C) 84 cm^2 (D) 96 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 84 cm^2** [0.5 Mark]

Given: $l+b+h=6\sqrt{3}$ and $\sqrt{l^2+b^2+h^2}=2\sqrt{6} \implies l^2+b^2+h^2=24$.

Formula / Theorem Used: $(l+b+h)^2=(l^2+b^2+h^2)+2(lb+bh+hl)$

$$\text{TSA}=2(lb+bh+hl)=(6\sqrt{3})^2-24=108-24=\boxed{\mathbf{84}\ \text{cm}^2}$$

Q5. **Assertion (A):** If the volumes of two spheres are in the ratio 343 : 1331, then their radii are in the ratio 7 : 11.
Reason (R): If the radii of two spheres are in the ratio 2 : 3, their surface areas are in the ratio 4 : 9. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** [0.5 Mark]

Verification: $\left(\frac{r_1}{r_2}\right)^3=\frac{343}{1331}=\left(\frac{7}{11}\right)^3 \implies \frac{r_1}{r_2}=\frac{7}{11}$ (Assertion (A) is true). Also $\left(\frac{2}{3}\right)^2=\frac{4}{9}$ (Reason (R) is true). Since (R) computes surface-area ratio for 2 : 3 rather than explaining volume-to-radius cube roots, (R) is not the correct explanation of (A).

SECTION B: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. What is the ratio of the volume of a cube to that of the largest sphere that can fit inside it? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Sphere Radius Inside Cube [1 Mark]

Let cube edge $= a$. The largest inscribed sphere has diameter $= a \implies r=\frac{a}{2}$.

Formula / Theorem Used: $V_{\text{cube}}=a^3$ and $V_{\text{sphere}}=\frac{4}{3}\pi r^3$

Step 2: Ratio Simplification [1 Mark]

$$\frac{V_{\text{cube}}}{V_{\text{sphere}}}=\frac{a^3}{\frac{4}{3}\pi\left(\frac{a}{2}\right)^3}=\frac{a^3}{\frac{\pi a^3}{6}}=\frac{6}{\pi}=\frac{6}{22/7}=\frac{42}{22}=\frac{21}{11} \implies \boxed{\mathbf{6}:\pi\ \text{(or } \mathbf{21}:\mathbf{11})}$$

Q7. What is the ratio of the volumes of a cylinder, a cone, and a sphere, if each has the same diameter $2r$ and same height $2r$? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Volume Formulas ($h=2r$) [1 Mark]

Formula / Theorem Used: $V_{\text{cyl}}=\pi r^2 h$, $V_{\text{cone}}=\frac{1}{3}\pi r^2 h$, $V_{\text{sphere}}=\frac{4}{3}\pi r^3$

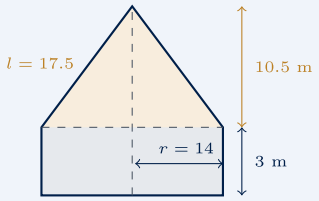
Step 2: Substituting $h=2r$ & Ratio [1 Mark]

$V_{\text{cyl}}=2\pi r^3$, $V_{\text{cone}}=\frac{2}{3}\pi r^3$, $V_{\text{sphere}}=\frac{4}{3}\pi r^3$.

$$\text{Ratio}=2:\frac{2}{3}:\frac{4}{3}=6:2:4=\boxed{\mathbf{3}:\mathbf{1}:\mathbf{2}}$$

Q8. A tent is of the shape of a right circular cylinder up to a height of 3 metres and then becomes a right circular cone with a maximum height of 13.5 metres above the ground. Calculate the cost of painting the inner side of the tent at the rate of Rs. 2 per m^2, if the radius of the base is 14 metres. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Dimensions & Slant Height [1 Mark]

$r = 14$ m, $h_{cyl} = 3$ m, $h_{cone} = 13.5 - 3 = 10.5$ m; $l = \sqrt{14^2 + 10.5^2} = \sqrt{306.25} = 17.5$ m.

Formula / Theorem Used: Inner SA $= 2\pi r h_{cyl} + \pi r l = \pi r(2h_{cyl} + l)$

Step 2: Area & Painting Cost [2 Marks]

SA $= \frac{22}{7} \times 14 \times (6 + 17.5) = 44 \times 23.5 = 1034$ m^2.

Cost $= 1034 \times 2 =$ **Rs. 2,068**.

Q9. A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a cylinder full of water of radius 60 cm and height 180 cm. Find the volume of water left in the cylinder. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Solid & Cylinder Volumes [1.5 Marks]

Common radius $r = 60$ cm; $h_{cone} = 120$ cm; $H_{cyl} = 180$ cm.

Formula / Theorem Used: $V_{left} = V_{cyl} - (V_{cone} + V_{hemi}) = \pi r^2 H_{cyl} - \frac{1}{3}\pi r^2(h_{cone} + 2r)$

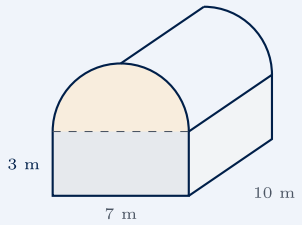
Step 2: Simplifying & Converting to m^3 [1.5 Marks]

$V_{left} = \pi(60)^2(180) - \frac{1}{3}\pi(60)^2(120 + 120) = 3600\pi(180 - 80) = 3,60,000\pi$ cm^3.

Converting ($1\ m^3 = 10^6\ cm^3$): $V_{left} = 0.36 \times \frac{22}{7} \approx$ **1.131 m^3** (360000π cm^3).

Q10. A godown building has a vertical cross-section parallel to the width side shaped as a rectangle of size 7 m × 3 m surmounted by a semi-circle of diameter 7 m. The internal length of the godown is 10 m.
(i) Find the internal volume of the godown building.
(ii) Find the total internal surface area of the building excluding the floor. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Given: Rectangle $w = 7$ m, $h = 3$ m; semi-circle $r = \frac{7}{2} = 3.5$ m; length $L = 10$ m.

Step 1: (i) Internal Volume [2.5 Marks]

Formula / Theorem Used: Volume = CS Area × $L = \left(wh + \frac{1}{2}\pi r^2\right) L$

CS Area $= (7 \times 3) + \frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = 21 + \frac{77}{4} = 21 + 19.25 = 40.25$ m^2.

Volume $= 40.25 \times 10 =$ **402.5 m^3**.

Step 2: (ii) Internal SA (No Floor) [2.5 Marks]

SA = 2 side walls ($2Lh$) + 2 ends (2 × CS) + curved roof ($\pi r L$)

$$\text{SA} = 2(10 \times 3) + 2(40.25) + \left(\frac{22}{7} \times 3.5 \times 10\right)$$

$$= 60 + 80.5 + 110 = \textbf{250.5 } m^2$$

Q11. Case Study: Heritage Architecture – The Great Stupa at Sanchi

The Great Stupa at Sanchi features a grand hemispherical brick dome surmounted by a square cuboidal harmika structure on top. The hemispherical dome has a base radius of 14 metres, while the cuboidal superstructure has dimensions 8 m × 6 m × 4 m. [4]

(i) Find the volume of the hemispherical dome of base radius 14 m. [1]

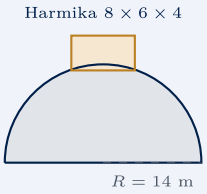
(ii) Find the volume of the cuboidal superstructure. [1]

(iii) Find the curved surface area of the hemispherical dome. [1]

(iv) Find the area of cloth required to cover the exposed hemispherical surface if the cuboidal base (8 m × 6 m) rests directly on it. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Dome radius $R = 14$ m.
Harmika cuboid: $l = 8$ m, $b = 6$ m, $h = 4$ m.



Formula / Theorem

Used: $V_{\text{hemi}} = \frac{2}{3}\pi R^3$, $V_{\text{cuboid}} = lbh$, $\text{CSA}_{\text{hemi}} = 2\pi R^2$

Sub-part (i): Volume of Hemispherical Dome [1 Mark]

$V_{\text{dome}} = \frac{2}{3} \times \frac{22}{7} \times 14 \times 14 \times 14 = \frac{44 \times 2 \times 196}{3} = \frac{17248}{3} = \boxed{\mathbf{5749\frac{1}{3}\ m^3}\ (\approx \mathbf{5749.33\ m^3})}$.

Sub-part (ii): Volume of Cuboidal Superstructure [1 Mark]

$V_{\text{harmika}} = 8 \times 6 \times 4 = \boxed{\mathbf{192\ m^3}}$.

Sub-part (iii): Curved Surface Area of Hemispherical Dome [1 Mark]

$\text{CSA} = 2 \times \frac{22}{7} \times 14 \times 14 = 44 \times 28 = \boxed{\mathbf{1232\ m^2}}$.

Sub-part (iv): Exposed Dome Surface Covered by Cloth [1 Mark]

Covered base footprint $= 8 \times 6 = 48\ \text{m}^2 \implies$ Cloth Area $= 1232 - 48 = \boxed{\mathbf{1184\ m^2}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★